

Collaborative Project Management Software

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Abstract

Project Management (PM) principles are rapidly changing due to business globalization and information technology (IT) advances which support distributed and virtual project teams. Traditional PM focuses on a single project at a single location [11] and is more concerned with project inputs and outputs than with the project work itself [42]. Traditional PM emphasizes scheduling, planning and tracking [11]. The PM paradigm has begun to shift due to the increasing number of distributed projects involving team members from different sites, organizations, and cultures [23]. Current and future PM will be more concerned with project work and processes, and collaboration will become essential for success. Academics and practitioners have begun to discuss project management in terms of collaboration; however, there is little agreement as to a definition or interpretation of collaboration.

One of the major changes in PM, over the last 25 years, has been the use of computerized tools and methods [17]. In the future, the most significant change in PM may be the use of collaboration [17]. In the article, we provide an explanation of different levels of collaboration, namely, communicative, collective, coordinated, and concerted. Then we describe prototype collaborative project management software (C-PMS) to support ‘concerted-level’ collaboration among project team members. We believe that C-PMS that supports all levels of collaboration, especially the concerted level, can increase the efficiency and effectiveness of project managers and team members.

1. Introduction

PM fundamentals are changing due to business [23] and technological forces [45]. Market globalization and

international mergers are increasing the need for partnerships across organizational, cultural, and national boundaries. IT advancements have made collaboration in distributed modes possible, technologies like Group Support Systems (GSS) and videoconferencing enable people to collaborate across dispersed geographical areas. The trend toward business globalization and the advancement of IT has given rise to new organizational forms called ‘virtual organizations’ [4, 11, 13]. Both traditional organizations and virtual organizations may employ ‘virtual projects’ involving team members from different geographical sites, organizations and cultures [27]. The challenge of virtual PM is supporting collaboration among the people working at different sites, different times or in different organizations.

Collaboration can consume a significant amount of project workers’ time. As more companies and industries move toward electronic commerce and digital business a large percentage of projects involve software development. “*Studies of co-located development teams in large (software) projects have shown that up to 70% of developer’s time is spent collaborating [22], and 85% of project costs can be accounted for by team activities [7]*” [14]. More traditional projects may also require fair amounts of collaboration. Helbrough [17] found that for a major business projects between 25% and 30% of total project man-days was spent for collaboration. Since collaboration is an important part of the PM, successful management of collaboration may increase productivity and decrease project costs.

PM must consider different project aspects. PM can be defined as ‘a special way of organizing, planning and controlling to create defined changes or products with predictable cost, time and quality’ [17]. PM thus involves managing financial issues, risks, resources, schedules, quality, and processes. Each of these areas must be

managed well for a project to be successful, however process management plays a more important role in a multi-site projects, because it involves task identification, task allocation, task tracking and reporting.

Good PM can provide team members and other stakeholders with a clear overview of who is doing what, and the status of a specific task. Visibility of project progress is a critical factor for project success [25]. Our Collaborative Project Management Software (C-PMS) provides collaboration for processes and activities as well as traditional tracking of finances, schedules, resources, and quality.

This paper is organized as follows: section 3 presents a literature review of PM from different perspectives; section 4 discusses different levels of collaborative PM and summarizes some commercial PM software functions in terms of collaborative support; section 5 presents our prototype Collaborative Project Management Software (C-PMS); section 6 explains a Framework for Research in Collaborative Project Management; section 7 presents a discussion and future research directions

2. Literature Review

We conducted our literature review in several fields: collaborative PM, organizational memory and knowledge management, collaborative engineering of processes and systems, and Group Support Systems, because work in collaborative PM is taking place in all of these areas. Each research area is focusing on different aspects of collaborative PM and therefore has something unique to offer.

2.1 Collaborative Project Management

Turner [41] defines a project as ‘*an endeavor in which human, material and financial resources are organized in a novel way, to undertake a unique scope of work, of given specification, with constraints of cost and time, so as to achieve beneficial change defined by quantitative and qualitative objectives.*’ The aim of PM is to produce the required deliverables within the constraints of time and budget. Without efficient and effective project management, projects risk schedule and budget overruns or even cancellation. 31% of new IS projects are canceled before completion at an estimated total cost of \$81 billion [12].

Project complexity is one of reason for project failures. Project complexity can be considered from at least three different perspectives: the number of projects; the number of physical sites; and number of organizations involved. Evaristo and Fenema [11] classified projects into seven

types based on the two dimensions of single versus multiple projects and single versus multiple sites (See figure 1).

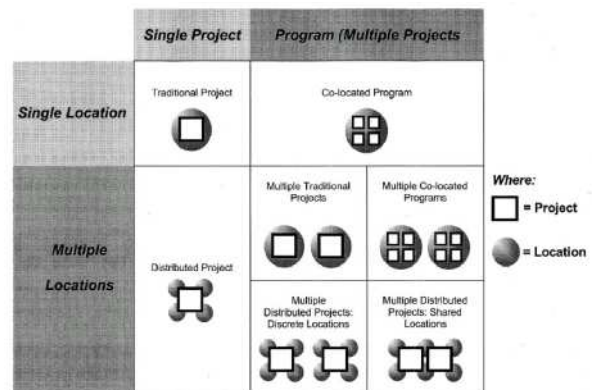


Figure 1. Evaristo and Fenema's Project Classification Scheme [11].

For traditional PM the key issue is scheduling [11], for multiple traditional projects, in addition to scheduling, there is a need to share resources to achieve global optimization. Evaristo and Fenema [11] point out that “*a critical difference between distributed projects and the prior programs or traditional projects of various types is related to the focus on the coordination mechanisms*”. In fact, coordination and collaboration are necessary and critical components for any project: coordination is within site for traditional projects, and across sites for distributed projects. Distributed projects with global perspectives have advantages and disadvantages. Advantages include: project has the right mix of personnel [1], cost-effective global sourcing, and shorter product development cycles by utilizing time zone differences [11]. One disadvantage is the high demand for communicative, cooperative and collaborative efforts among team members [11]. Inefficient collaborative efforts may offset the time zone difference gains [14].

One major change of PM in the last 25 years is the use of computerized project planning and control methods [17]. PM software packages are available to provide Gantt charts, PERT diagrams, resource histograms, and project status tracking. There are many factors to manage for a project, and project managers cannot pay equal attention to each factor, therefore they prioritize factors and decide which ones to pay the most attention to. Evaristo and Fenema [11] point out that traditional PM emphasizes management of a single project at a single location, and the key issue is scheduling. This might be why people often think about Gantt Charts and PERT diagrams when they think about PM. People manage project inputs, resources, delivery of products, and people, but seldom

manage the work or process of the project [42]. If PM concerns mainly inputs and outputs of the development effort, the process of the project may remain a 'black box'. There are many chances that things can go wrong in the 'black box', and people will not be aware of whether the project is a failure or success until the last minute. Some PM tools keep track of the task status to increase the visibility of the process and gain more control of the project. However, task status tracking is not sufficient. For example, if people are doing the right things, and not sharing or archiving the knowledge used in the process, they won't be able to make the process repeatable by other people or by next project. Process and content support is necessary for successful PM.

In traditional PM, progress tracking might be sufficient to manage the process. However, progress tracking is insufficient for distributed projects due to the dynamics and complexity of the project. People in different locations or organizations need to know not only the status of the most current work, but also the actual work itself. Careful coordination of subtask allocation, interdependence, and integration is needed on a daily basis. A virtual project is temporary in nature, it is formed for a specific project and dissolved at the end of the project [27], [37]. It is very important to keep the process of the project in the organizational memory so that successful process can be repeated for the future project, and problematic process can be avoided.

In summarizing the reason of project failure, Helbrough [17] states that poor collaboration is often cited as the reason, and he envisions that collaborative technology may be the most significant change in the way projects are managed in the future. Helbrough does not provide a definition of 'collaboration', however he specifies that '*collaboration means meeting*'. From his point of view, collaboration is meeting, the effectiveness of collaboration can be achieved by using GSS(Group Support Systems). We agree with him that collaboration might be the most important dimension of project management, however his interpretation of collaboration is limited, though GSS may be an important component of collaborative PM. From the literature review, we realized that different people have different understandings of collaboration. Some companies claim they provide collaborative PM software and/or services.

A closer look at the explanation of the software and services will illustrate the variations in interpretation of collaboration. For example, Netmosphere has a collaborative project management tool named "ActionPlan", which allows team members to access a central information store via an Internet interface.

Netmosphere CEO Kevin Nickels interpretation of collaboration is different from that of Helbrough. Nickels' understanding of collaboration can be illustrated by his statement - "*most of the traditional views of project management are not collaborative. I've seen charts where the project manager is drawn as the center of the universe. The executives are on the outside ...*". Netmosphere's tool allows everyone related to the project to participate. "*Everybody is on the team, and nobody is the overseer*" [21]. It seems that collaborative is the same as participative for Nickels. Other companies simply regard collaboration of PM as putting project management on the web so that project management can be done anytime, from anywhere. The variation of interpretation of collaboration is confusing and even misleading.

Even if people have varied interpretation of the collaboration in PM, they agree that collaboration is a critical component in new PM paradigm. Increasing numbers of distributed projects have led to a shift in the PM paradigm. If traditional PM manages inputs and outputs of one project, and the key is scheduling, then current PM focuses on managing the work or process of the project and the key is collaboration. In order to achieve the effectiveness and efficiency of collaboration in PM, it is necessary to define collaboration in the first place, and then understand the system functions in terms of collaboration. We will explain what we mean by collaboration in next section.

Organizational Memory and Knowledge Management

Literature on Organizational Memory (OM) is relevant to collaborative PM because it provides a team repository that is lacking in traditional PM, but required for collaborative PM. Organizational memory (OM) is defined as "*the means by which knowledge from the past is brought to bear on present activities, thus resulting in higher or lower levels of organizational responsiveness*" [15]. The processes of OM are acquisition, maintenance, retention, and retrieval [39]. Acquisition gathers information from all available sources, the information should be new – not exist in the maintenance process (database). Maintenance is to keep the information. Retention is to facilitate people to organize and process information effectively. Retrieval provides information to support decision making and problem solving.

Baek and Liebowitz [1] have done some research in web-based knowledge repository for multimedia design teams. The result indicates that knowledge repository creates a promising environments for these teams, since knowledge repository allows people exchange knowledge and be aware of the existing knowledge by sharing knowledge. Organizational memory is closely related to knowledge

management (KM) in terms of creating mechanisms to keep knowledge in a persistent format. Knowledge can be classified as explicit and tacit knowledge. Different activities transform knowledge from one category to the other: combination is to combine explicit knowledge to produce explicit knowledge; socialization is to transfer tacit knowledge into tacit knowledge; internalization is to transform explicit knowledge into tacit one; externalization is to turn tacit knowledge into explicit knowledge [28].

McQueen ([26]) summarized that there are two major research themes in KM: *“the first theme centers on how information technology can support both the capture of tacit knowledge held by individuals and its conversion to accessible explicit forms (often through discussion repositories using groupware such as Lotus Notes). The second theme involves the use of information technology for electronic publishing of corporate documents on an intranet, which will support the search for information by knowledge workers in the organization”*. The KM capacities of capturing tacit knowledge, turning it into explicit knowledge and keep it as persistent object for later use are important, PM software need to have such a KM component to keep knowledge for a current and future reference.

Group Support Systems

Literature on Group Support Systems (GSS) is relevant because it can support all stages of a collaborative PM process. Although little specific GSS research has looked at PM on the whole, several researchers have studied parts of the process. PM process components which have been studied with GSS include: strategic and project planning [8, 24, 31, 36, 43] requirements elicitation [5, 6, 18, 35, 33, 34]; and software inspections [44]. van Genuchten et al. [44] also explored the use of GSS in primary work processes. PM is for many a primary work process that GSS could support.

Collaborative Engineering of Processes and Systems

The literature review of this area illustrates types of communications need to be supported by the collaborative PM tools. Munkvold and Evaristo [27] states that a virtual project needs to have a collaborative infrastructure to support its implementation. They develop a state model of collaborative infrastructure implementation: first stage mainly deals with issues of hardware, in this stage, collaboration hardware is installed and tested; second stage deals with software readiness, collaborated software is installed, tested and available to users; third stage relates to available guidelines of collaboration between different sites, for example, who should hold of the

“editable” version of the documents and who can only access the “read only” version.

The literature discusses three major project types: 1) a single project conducted at single location, 2) a single project conducted at distributed locations, 3) multiple projects conducted at multiple locations. For type 1 projects, communication is conducted asynchronously via email and other communication tools. Other technologies used include: technologies for supporting coordination (e.g. workflow management, calendaring, and scheduling), and information sharing (e.g. document management systems and knowledge repositories). For type 2 projects, the technical infrastructure needs to accommodate telecommunication capabilities (internet connection and alternatives), and interoperability of different computer platforms. Type 3 projects create special needs for a collaborative infrastructure, because project aspects such as synchronicity, culture, structure, policies and standards, the infrastructure need to support prioritization schemes, cross-site and cross-project calendaring and other management tools. Ideally a collaborative PM software tool should support at least type 2 projects.

Collaboration in Project Management

PM fundamentals have changed, not only in terms of the projects, but also in terms of computing environments [45]. Two decades ago, mainframe computers were used for PM, with centralized data storage and access via terminals. Resources were shared and access was controlled. Sharing central resources can create performance bottlenecks and slow response times. Using PCs for PM can improve response times. However three drawbacks are associated with PC use: data synchronization, communication, and interoperability of different platforms.

New technologies have emerged to cope with PM needs. People use central data repositories to solve the problem of synchronization: data is physically replicated on many different servers, all synchronized and updated so that everybody appears to be working with one data source. Open Database Connectivity (ODBC) and Object Linking and Embedding (OLE) automation are two technologies that can overcome the problems of mixed-platform environments. ODBC allows applications to connect Structured Query Language (SQL) database systems and enables applications to communicate with project data no matter where or how it is stored. OLE automation technology provides applications with capabilities to use individual components of other applications. For example, one application can ask another to open the file without

knowing the file storage format or location [45]. Internet and intranet offers new forms of communication between project participants. Advances in computing environments make collaboration for distributed project members feasible.

Levels of Collaboration

A number of authors [2, 9, 10, 16, 19, 20, 38] have discussed various types of collaborative support and Nunamaker co-authors [29, 30, 32] propose a five level hierarchical framework for collaboration: Communicative; Collective; Coordinative; and Concerted (See Table 1). It is the 'concerted' level of collaboration that will add real value to project Management.

Levels	Nature of Efforts	Team Productivity	Process/Tasks	Applications	Metaphors
Communicative	Dialog and Common Understanding Dominated or Egalitarian Information Sharing Non-Task Oriented	Minimal	Informal / semi-formal communication Conversational Chaotic Unstructured Uncoordinated	Email Chat Notes Data bases News Groups Computer-mediated Communication	Huddle Hallway Meeting Coffee Clutch
Collective	Individual Separate Piece-Meal Tasks	Sum of Individual performances	Individualized (Start-end) Minimal Integration	Word Processing Spreadsheets Graphics	Sprinters
Cooperative	Group-Based Uncoordinated	Sum of individual Performances	Ad hoc Separate tasks toward common goal Final integration	Net Meeting Application Sharing	Hockey Soccer
Coordinated	Sequential Team Precedential	Sum of Sequential Individual Performances	Chronological Step-by-step Ordered Hand-offs Work-Flow Oriented Progressive Integration	Work flow	Relay Work Flow
Concerted	Mutual Communal	Sum of individual, Ad hoc Team and Coordinated Team Performances	Jointly Shared Synchronized Continuously Integrative Simultaneous Flexible Repeatable Customizable	Group Support Systems Computer-Supported Collaborative Work	Crew Tug-of-War Group Writing Group Problem - sSolving

Table 1. Levels of Collaboration [29, 30, 32]

Collaborative PM software can be classified into four levels, and the higher level software has all the capabilities of the lower level software. From the above comparison of different collaborative PM tool, we can see that current tools only support the lower levels of collaboration.

Communicative level software is web-based and only provides a central knowledge repository for file storage. Every project member can access the information anytime, from anywhere. People exchange information via the repository, and everyone works on his/her own. Baek and Liebowitz [1] has illustrated knowledge repository is good for project members to share knowledge. However, it does not support interaction among team members, and the interaction is very important for team cooperation success.

Collective level software provides a repository for information storage, tasks, timelines, and resource tracking. People are working toward a common goal and can check the updated work of other people. The integration of member's work is minimal.

Coordinate level software allows users to create, share, review, and redline project documents, check calendars, coordinate schedules, and review tasks. It allows people to

coauthoring a document in a sequential manner. level software has the capabilities of collective level software and features of group calendar, and Net meeting or online discussion forum. People work under the common goal and interact and cooperate with each other.

Concerted level software allows project members to coauthor a document in a parallel manner, and the software supports synchronous and asynchronous group problem solving and decision making.

The lower levels of collaboration focus mainly on information sharing. Information sharing is important, but information sharing alone is not sufficient. Project members at different sites need to do their tasks depending on other people's job. The interaction between project members is essential, and interaction supporting is an important component for collaborative PM tool. Interactions include negotiation of the goals, task allocations, scheduling, resources allocation, co-work on the same document or same task. Without effective interaction support, project members can work very hard and find they are working toward a misunderstanding goal.

The concerted level of collaboration really offers effective and efficient support for managing the complexities of a distributed project. We propose Collaborative Project Management Software (C-PMS), a prototype of concerted level project management software, in next section.

At present not very many companies are practicing systematic collaborative PM. There are several reasons: first, people don't have the tools or the mechanisms to do Collaborative PM. Surubhotla [40] points out that "*most (software) development organizations have only superficially instituted the practices and processes involved with serious project management*", despite the fact that organizations emphasized the use of standardized project management techniques as a means to improve development productivity. The reason may be that available software does not yet fully support the processes in PM. Second; the user learning curve for PM software may be an obstacle. Third, companies have time and

money constraints to implement the PM software. Fourth, lack of management commitment to PM practices may be a problem. [3] The fourth reason can be interpreted as lack of motivation to practice PM. There may be other reasons for not conducting collaborative PM in organizations. In this paper, we address the first issue: lack of tools. We propose prototype environment collaborative project management.

Present CPM Tools

Many companies claim they have collaborative project management tools, and these tools support different levels of collaboration. Figure 2 provides a brief view for some tools: company name, PM tool name, project types the tool supports, and the collaboration level the tool supports. Appendix A lists the detailed description of the PM tools, such as the company's interpretation of collaboration project management, name of tools, and the major functions of the tool.

Company Name	PM Tool Name	Category of project	Collaboration level
Rational.Com	Rational Soda	Software Project	Collective
Onproject.Com	Onproject	Business Project	Coordinated
Citadon.Com	Projectnet Projectnet Process	Project Of Engineering, Building, And Real Estate	Coordinated
Surdex.Com	CPMS (Collaborative Project Management System)	Photography And Mapping Services	Communicative
Viecon.Com	Projectbank Projectwise	Engineering	Coordinated
Microsoft.Com	Microsoft Project 2000	Any Kind Of Project	Coordinated
Inovie.Com	Team Center 4.5	Any Kind Of Project	Coordinated

Table 2. Collaboration Level of Collaborative PM Software

Prototype Collaborative Project Management Software (CPMS)

Our proposed prototype CPMS provides teams with the capability to manage multiple projects involving geographically dispersed teams. The CPMS will have the ten modules listed below. Figure 2 illustrates eight of the modules.

- 1) Project Specification Overview – stores project specification: name, goal, sponsors, customers, leaders, duration, involving locations and collaboration policies.
- 2) Accounting and Cost Management module – tracks of the cost as project progresses.
- 3) Administration and security module – Allows leaders/system administrators to set up new project and new users, and manage them.
- 4) Content Management module – Provides central knowledge repository for the project. It stores all kinds of files, documents and meeting notes. Document management includes user access control, check-in, check-out functionality, status-

and revision management, management of different file types, scanning, and archiving.

- 5) Resources management module – Manages project resources such as people, equipments, and services.
- 6) Process Management module – Manages work flow, process templates, project phases, tasks, schedules for each task, and delegation of tasks to project members. Researchers in Center for Management of Information (CMI) have developed Project Tracker, which is a software project management system prototype. This prototype is a web-based project management tool, it can be modified to support content, resources, and process management.
- 7) Collaboration module – Facilitates the collaborations between project members within and across project sites. Project members can communicate via synchronous and asynchronous pattern. Researchers in Center for Management of Information have developed a Group Support System named Collaboratus, it can be used to

- conduct group writing and on-line group meeting.
- 8) Data Visualization module – Allows users to view project status, cost and resources in an intuitive way.
 - 9) Graphical User Interface module – Provides the user interface to interact with the system.
 - 10) Database module – Stores all data for the system.

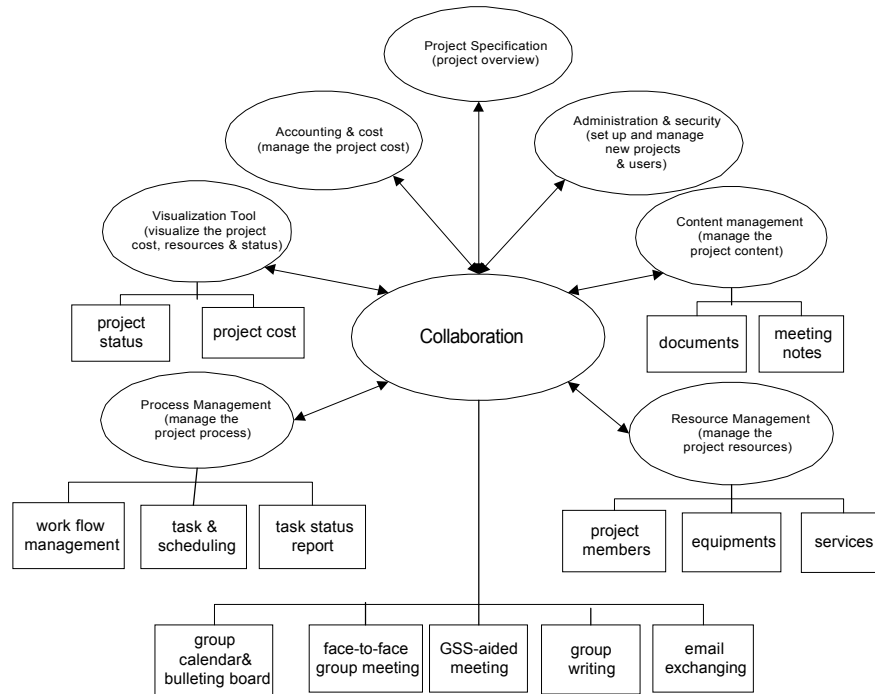


Figure 2. System Modules for Prototype Collaborative Project Management Software

Framework for Research in Collaborative Project Management

We will evaluate our prototype with a mixed method design both quantitatively and qualitatively. We will conduct the usability study first. The usability study will test whether the system is usable: Is the user interface is intuitive? Is the user's learning curve is steep? Is the system's response time in the reasonable limit? Is there any important functionality not included in the system? Is the system stable enough for practical use? We will also conduct experiments by using students as subjects. The aim of experiment is to find out the usefulness and acceptance of the system, and to identify the features need to be improved, revised, added and dropped. Following the experiments, we will do case study. Center for Management Information at the University of Arizona has several projects going, some of the projects involve distributed sites. The researchers manage their projects by using the prototype system. While experiments will answer the questions of "What", case study is good at getting the answers of "Why". We will test whether our

conceptual model of collaborative PM is feasible and whether collaborative PM increases the project management effectiveness and efficiency.

7. Discussion

In this paper, we have discussed the PM paradigm, its essentials, and the key feature of collaboration. PM fundamentals have changed due to the business globalization and advances of computing environments. Traditional PM focuses on a single project located in a single location, the key issue is scheduling and planning. In the past 25 year, the revolutionary change of PM is the computerized PM. However, the computerized PM has focused on the automation of scheduling and planning, and adopted single-user or single-project perspective. The next big change of PM will be collaboration [17]. More organizations now employ virtual project teams to get the right mix of personnel, cost-effective global sourcing, shorter product development cycle, and time gains by utilizing time zone differences. The distributed project needs strong support of collaboration among project

members. The ineffective collaboration can offset the benefits of virtual project. There are different levels of collaboration. Most of the commercial PM software provides lower level collaboration support. We think the ideal collaborative support should be at "concerted" level.

We have proposed a prototype collaborative project management software (CPMS). The prototype allows project teams to manage multiple projects involving one or more project sites in a concerted collaborative fashion. The system provides a persistent knowledge repository to store the knowledge and information used and produced by the project members. It also allows users to communicate in both synchronous and asynchronous modes: team members can conduct on-line group meetings, software inspections, or send email. The meeting notes and emails will be stored in the system for future reference. Project members can also coauthor a document either at the same time or different times. This is expected to decrease turn-around-time for documents and increases both efficiency and effectiveness of individual project team members and project teams on the whole.

Our future research direction include implementing our prototype CPMS and performing a case study to analyze how it affects both PM process and outcomes from both qualitative and quantitative perspectives.

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